

Lemma 3.5

distinct

- $i, k \in \{1, 2, 3\}$

let i unique

s.t. $\{i, j, k\}$

$= \{1, 2, 3\}$

- supp $v_i \notin V(C)$

- Cycle C

with $e_j \in C$

$e_k \in C$

s.t.

$\forall v \in V(C)$

$\kappa(v) \geq \kappa(v_0)$

- $v_n \in V(C)$

with $\forall v \in V(C)$

$\kappa(v) < \kappa(v_n)$



$\kappa(v) \in [\kappa(v_0), \kappa(v_n)]$

- h'_i comp of $h - V(C)$

with $v_i \in h'_i$

$\forall v \in h'_i$
 $\kappa(v_0) < \kappa(v) < \kappa(v_n)$

what's the meaning behind this
 a cycle with no support
 any blue from $h - V(C)$

Proof)

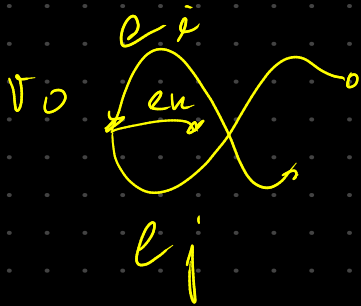
P_j, P_k



suppose $\kappa(v_i) > \kappa(v_k)$

let consider $(i, j, k) = (1, 2, 3)$

v_j above e_i v_k below e_i



At v_k both paths from e_j and from e_k will pass from the same vertex

AND

$\forall v \in P_j$ are above e_i

AND

$\forall v \in P_k$ are below e_i

$\Downarrow$

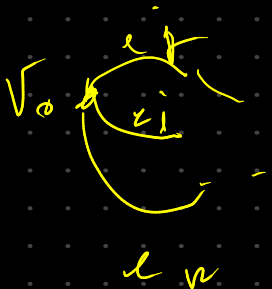
we have a conflict in v_k .

$(1, 2, 3)$

for v_1 $\kappa(v_0) < \kappa(v_2) < \kappa(v_3)$

* Case $(i, j, k) = (1, 3, 2)$ is the same
 * Case with $i=2$ is symmetric.

* sup. that $i=3$ w.l.g. $(j, k) = (1, 2)$
 $(i, j, k) = (3, 1, 2)$


 $\Rightarrow \forall v \in P_i - v_0 \quad v \text{ is above } e_i$
 $\left\{ \begin{array}{l} \forall v \in P_i - v_0 \quad v \text{ is below } e_i \end{array} \right.$

contr. since v_e is above and below e_i

we may assume
 $\Rightarrow \kappa(v_0) < \kappa(v_i) < \kappa(v_R)$ Same

as the proof of 3, 4 or i below P_i or X
 $\rightarrow$ so the path P_i is both above P_i and below P_n

* sup. $\exists P_i' \in h - v(L)$
 $v_i \rightarrow v' \quad / \quad \kappa(v') < \kappa(v_0)$
 or
 $\kappa(v') > \kappa(v_R)$

Th 3.1)

h has a

h -monotone

drawing

with
indep. edges.

crosses evenly

(Prod) Case 1

h has a
 π -monotone

embedding

(same vertex locations)

v_1, v_2, v_3 diff comps
of $h_0 - v_0$

$\forall i \in \{1, 2\}$, consider comp $h_0 - v_0$ that
contains v_i .

let v'_i be largest π -valued vertex in

pick $i, j, k \hookrightarrow \begin{cases} \{1, 2, 3\} = \{1, 2, 3\} \\ \pi(v'_i) < \pi(v'_k) \\ \pi(v'_i) < \pi(v'_j) \end{cases}$

let $\pi_k = \pi(v'_i)$

(Lemma 3.4) $\Rightarrow$ we have
 h'_i, h'_j, h'_k

$\forall l \in \{1, 2, 3\}$
 $\Rightarrow$ for all
comps h'_l have no
neighb. (in h) to
the left of $\pi(v_0)$

then we can use lemma 3.3 (ii)

$$\begin{cases} a = v_0 \\ H = h_i \end{cases} \rightarrow \begin{cases} h' = \{r_i\} \Rightarrow v'_i = v_i \\ \kappa(r_i) > \kappa(r_j) \\ \kappa(r_i) > \kappa(r_a) \end{cases} \quad (6)$$

$\Rightarrow h_j, h_k$ are non-empty (we can use lemma 3.4)

$\Rightarrow$ for all h_l $\forall l \in \{1, 2, 3\}$ have no
neigh. (in h) to
the left of $\kappa(r_0)$

(contradiction) lemma 3.3 (ii)

h has no connected subgraph $H' \neq \emptyset$ so that

$$\kappa(a) < \kappa(v) < \kappa(b) \quad \{h_j, h_k\}, \quad \forall v \in h_j \cup h_k$$

$$a = v_0 \in N(H') \neq \{a\} \text{ and } \kappa(v) > \kappa(b)$$

$$\forall v \in N(H') - \{a\}$$

if there is no Neigh. of h_j on the left
 of x_0 it means either

$$\begin{aligned} & N(H') = \{a\} \\ & N(H') \neq \{a\} \\ & \{v \in N(H') - \{a\} : x(v) > x(a)\} \end{aligned}$$

breaks
 the minimal
 example



if we are not in case 1, let x_L be smallest

s.t. h' induced by $\{v \in V(h) : x(v_0) < x(v) \leq x_L\}$

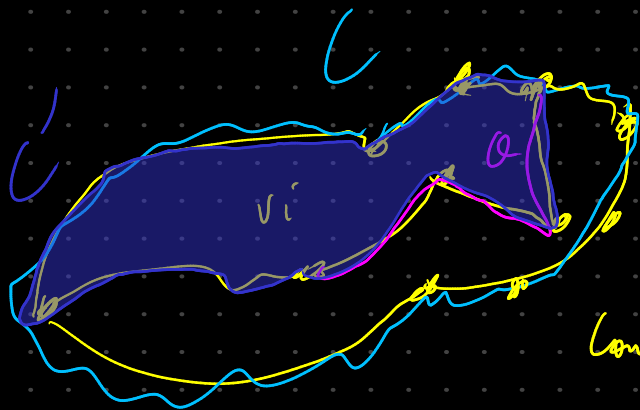
has a comp that contains at least two vertices

of $v_1, v_2, v_3 \Rightarrow$ cycle C containing e_j, e_k
 distinct $(j, k) \in \{1, 2, 3\}$, vertex v_L with $x(v_L) = x_L$.

let $x(v_0) < x(v) \leq x(v_L) \forall v \in V(C)$.

if $v_L \in \{e_1, e_2, e_3\}$ we may assume that

C contains v_L .



let $\forall v \in C'$
 $\kappa(v) \leq \kappa(v) < \kappa(v_L)$

contradiction on the choice
 of v_L

there is $\nexists$

Q from v_i to C

with $\forall v \in Q \quad \kappa(v) < \kappa(v) < \kappa(v_L)$

we can find Q without having Q disjoint
 from $C / \{v_L\}$

which also
 means

the 2

left cases

$v_i \notin C$
 because
 $v_i \neq v_L$



not possible
 by lemma 3.5

let v_i' largest κ -value
 of h_i'

let $\kappa_2 = \kappa(v_i')$

h_i, h_j, h_k | if we suppose $x_L = x(r_i)$
 then we cut the cycle
 so we have disjoint comp h_i, h_j, h_k again

$$\begin{cases} x(r_i') = x_R > x_L \\ x(r_i') > x(r_i) > x_R \\ x(r_i') > x(r_i) > x_R \\ > x_L \end{cases}$$

h_i has no N left of $x(r_i)$ i.e. $\{1, 2, 3\}$

h_i not adj to v_L so $h_i' = \{r_i'\}$



Lemma 3.3 (iii) $H = h_i' \quad V(H) - (V(H) \cap \{2, 4\}) \neq \emptyset$

3.3 (iii)

no connected subgraph
 with H

contradiction

(we still have at least v_j and v_k which $\notin H$.)

Lemma 3 - let h be a level graph

- P be a Hanani-Tutte drawing of h

$\Rightarrow S(\bar{h})$ is satisfiable

Proof

satisfies

all clauses of $S(\bar{h})$

object

use P to define a t.n. ϕ of $S(\bar{h})$.

* let $\bar{I}$ be the drawing of $\bar{h}$ $\searrow$ I

* let ψ t.n. $\searrow$ I

let $u'', w'' \in V(\bar{h})$ with $L(u'') = L(w'')$

$u'' \in V(h)$

w.l.g.

$$\begin{aligned} \phi(u'', w'') \\ = \psi(u'', w'') \end{aligned}$$

Lemma 2

$$\begin{aligned} \phi(u'', w'') \\ = \phi(u', w') \end{aligned}$$

indep

adjacent

$$\begin{aligned} \phi(u'', w'') \\ = \psi(v', x') \end{aligned}$$

or

$$\begin{aligned} \phi(u'', w'') \\ = \phi(u', w') \end{aligned}$$

$\Rightarrow$ any internal pair of edges (indep - adjacent)

$$(u'', v''), (w'', x'') \in E(\bar{H})$$

$$\text{that } \phi(u'' w'') = \phi(v'' x'')$$

Lemma 4

H is a level graph. Then $\overline{S(H)}$ is satisfiable

$$\Leftrightarrow S(\bar{H}^+) \text{ is satisfiable}$$

$$\bar{H}^+ = H^+$$

what's $\bar{H}^+$?

Proof

Let ϕ a satisfying t.a. of for $S(H)$

$$O: V(\bar{H}^+) \rightarrow V(H)$$

O gives originth of a vertex in $\bar{H}^+$

$$\forall v \in V(\bar{H}^+)$$

$$\exists u \in V(\bar{H}^+)$$

$$\exists w \in V(H)$$

v is part of a stretch edge

$\downarrow$
 u is
 a sub-
 division
 vertex

$$e(w)$$

$$O(v) = w \text{ of } \bar{H}^+$$

$\forall$ other $v \in V(\bar{H}^+)$ is a subdivision vertex of an edge (x, x') of $\bar{H}^+$ that is not a stretch edge.

for what's left? we map

v to vertex $w \in \{x, x'\}$ with

$$L(l(v)) = L(l(w))$$

the vertices
that are on the
same level of an
edge of G^*

so for $x, y \in V(T^+)$ with $O(x) \neq O(y)$

$$\text{we set } f^+(xy) = f(O(x)O(y))$$

If $O(x) = O(y)$



subdivide vertices of non-stretch
edges e_x, e_y in T^* .